Unite2:Assignment 5.py

Code:

```python
def lcs(X, Y):
    m = len(X)
    n = len(Y)

    dp = [[0] * (n + 1) for _ in range(m + 1)]

    # Build DP table
    for i in range(1, m + 1):
        for j in range(1, n + 1):
            if X[i - 1] == Y[j - 1]:
                dp[i][j] = dp[i - 1][j - 1] + 1
            else:
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1])

    # Find the LCS
    i = m
    j = n
    result = ""

    while i > 0 and j > 0:
        if X[i - 1] == Y[j - 1]:
            result = X[i - 1] + result
            i -= 1
            j -= 1
        elif dp[i - 1][j] > dp[i][j - 1]:
            i -= 1
        else:
            j -= 1

    return result


X = "AGGTAB"
Y = "GXTXAYB"
```

```
answer = lcs(X, Y)

print("Longest Common Subsequence:", answer)
print("Length:", len(answer))



--------output---------
Longest Common Subsequence: GTAB
Length: 4
```